SatQuery Flood Impact

Technical report | IIT Mandi Track B

1. Objective and scope

The isolated local application at port 8770 estimates flood-related candidates, infrastructure exposure and possible access disruption. The original SatQuery remains on port 8767. This prototype is not a validated emergency response system. The selected AI component is an English/Nepali evidence copilot; no trained flood-segmentation performance is claimed.

2. Inputs and provenance

The Nepal reference case uses matched Sentinel-1D acquisitions from 16/28 August 2026, relative orbit 85 ascending. Copernicus Sentinel Hub processes GRD with terrain correction and Copernicus DEM30; processing requests and output hashes are retained. Optical observations are Sentinel-2 L2A. Historical OSM is acquired with snapshot timestamps and original road versions. EMSR927 damage labels are evaluation-only and are excluded from prediction and threshold selection.

3. Processing

Scene identity, VV/VH bands, units, grids, orbit and dates are checked. Linear gamma0 is converted to dB; nonfinite, nonpositive, shadow, out-of-range incidence and NoData pixels are masked. Water candidates require post-event VV below -17 dB, pre-event VV not below that threshold, and a drop of at least 3 dB. These are uncalibrated defaults. Two-date water is not proof of permanent water.

The debris proxy requires vegetation-to-bare-ground optical change with a SAR change. Cloud-invalid pixels are excluded. It is not a trained classifier. Automatic acquisition examines up to three nearby optical dates on each side within 14 days and selects the greatest clear coverage; this bounded search may miss a better scene.

4. Exposure and access

Geometries are intersected in a projected metre CRS. Building counts, road lengths and bridge features retain source IDs. Historical OSM node IDs define junctions, so crossing lines are not automatically connected. Directed road access is compared before/after assumed candidate disruptions. Nearest-node settlement/hospital snapping is limited to 200 m; turn restrictions are not modelled. Network boundaries and unobserved roads retain uncertainty. No confirmed closure or isolation is inferred from the current Nepal result.

5. Interface and AI

Workspace exposes area/date inputs, background progress, saved runs, imagery, layers, downloads and questions. Evidence separates software verification from diagnostic reference comparisons. Qwen3-0.6B selects a topic; bilingual templates insert only saved calculations and source pointers. Routing scored 9/12 development questions before guardrails. This is not independent testing or unrestricted reasoning. Users can override topics; Nepali wording needs human review. Casualty/evacuation claims are unsupported.

6. Evaluation

The official AOI frozen run reports 36,100 m2 water candidates, one intersected building, 87.40 m intersected road/bridge length and two bridge features. Debris candidates cover 212,300 m2. One-AOI comparison with EMSR927 landslides yields IoU 0.3279, F1 0.4939, precision 0.9948 and recall 0.3285. This diagnostic has class, time and resolution differences and low recall; no threshold changes were made after scoring. It cannot establish generalization. Water has no compatible accuracy score.

Software checks cover grid/orbit/date contracts, graph cases, exposure, tiled pixelwise equivalence, frozen-evidence integrity and job-state failures. The current regression suite has 69 passing tests. Tests do not prove real road closures or prediction accuracy. Full arrays remain in memory; automatic requests are bounded to one million 10 m pixels and 2500 pixels per side.

7. Reproducibility and current acceptance

Run IDs cannot overwrite prior evidence. Outputs include GeoTIFF, GeoJSON, feature CSVs, calculation JSON, reports and SHA-256 hashes. Provider credentials stay in ignored local configuration. The CLI water/exposure and automatic topology/optical stages have completed real runs. A fresh integrated Workspace API job completed with SAR, optical/debris proxy, historical topology and frozen outputs. Browser visual verification, resource measurements, Nepali review and demo recording remain pending at report generation. See the current machine verification records for subsequent changes.

8. Attribution, reuse and limitations

Sources: Copernicus Sentinel and Copernicus DEM; OpenStreetMap contributors (ODbL) through HeiGIT/ohsome; Copernicus EMS reference products (checking only); Qwen/Qwen3-0.6B, pinned revision c1899de289a04d12100db370d81485cdf75e47ca. The visual foundation derives from the existing SatQuery application; preserve its MIT notice. Organizer permission for existing-code reuse was not established and must not be implied. No repository publication or submission has been performed.

Official specification source: multimodal-ai-hackathon-2026-7.devpost.com; local audit: docs/official-spec-audit.md. Model, imagery and OSM request provenance accompany the local evidence. Full upstream distribution terms must be preserved when publishing.